Topic 25:
Quantum Physics

25.1 Energy of a photon

25.2 Photoelectric emission of electrons

25.3 Wave-particle duality

25.4 Energy levels in atoms & line spectra

What is Quantum Physics?

http://www.youtube.com/watch?v=XEZtw1yt8Kc

Quantum Physics

Quantum physics:

- the study of the behaviour of matter and energy at the molecular, atomic, nuclear and smaller microscopic levels.

- In the early 20th century, it was discovered that the laws that govern macroscopic objects do not function the same in such small realms.

What is Quantum

"Quantum"

- comes from the Latin meaning "how much."
- It refers to the discrete units of matter and energy that are predicted by and observed in quantum physics.
- Even space and time, which appear to be extremely continuous, have smallest possible values.

Who Develop Quantum Physics

- As scientists gained the technology to measure with greater precision, strange phenomena was observed.

- The birth of quantum physics is attributed to Max Planck's 1900 paper on blackbody radiation.

- Development of the field was done by Max Planck, Albert Einstein, Niels Bohr, Werner Heisenberg, Erwin Schroedinger, and many others.

Particulate Nature of E/M Radiation

- In 1900, Max Planck, a German physicists suggested that the electromagnetic waves emitted from a blackbody (a perfect absorber and emitter of radiation) was quantized.

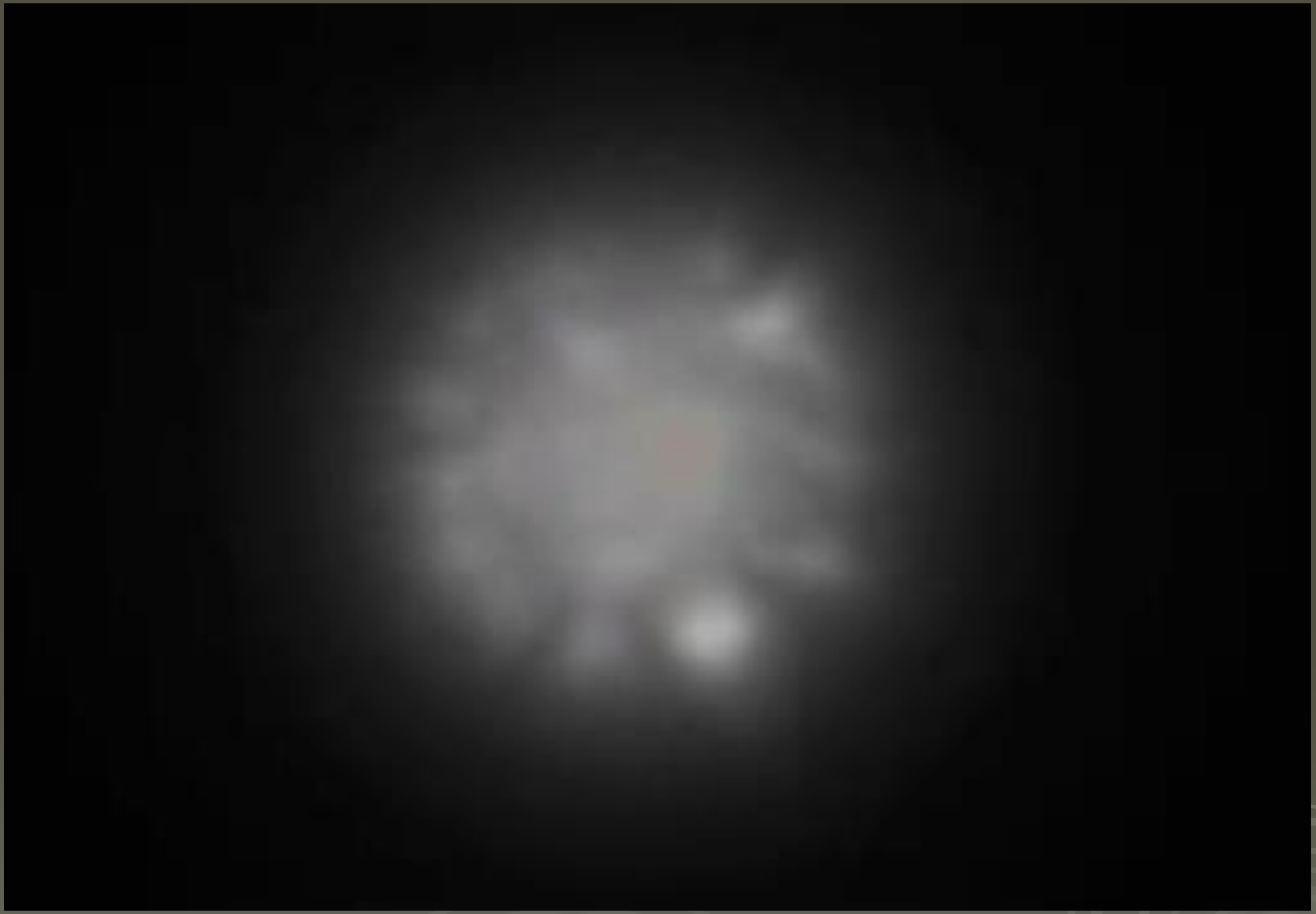
- This means that the energy emitted is not continuous, but instead consists of discrete amount or packets called quantas.

- In 1905, Einstein extended Planck's idea and postulated that light is emitted in packets (quantas or photons) and remains in packets till absorbed.

- This idea of quantization of electromagnetic waves into packets of energy called photons suggests a particulate nature of electromagnetic radiation.

The Particulate Nature

The Photon



The photon is a quantum or packet of energy of an electromagnetic radiation.

Energy of a Photon

$$E = hf$$

OR

$$E = \frac{hc}{\lambda}$$

Where:

h is Planck's constant = 6.63×10^{-34} Js

f is the frequency of the electromagnetic wave

λ Is the wavelength of the electromagnetic wave

c is the speed of light in vacuum = 3.00×10^{8} ms^{-1}

Example 1

Find the frequency and energy of a photon of wavelength 500 nm.

Solution:

The frequency of the photon is

$$f = \frac{c}{\lambda}$$

$$= \frac{3.00 \times 10^{8}}{500 \times 10^{-9}}$$

$$= 6.00 \times 10^{14} \text{ Hz}$$

Thus, the energy of the photon is

$$E = hf$$

$$= (6.63 \times 10^{-34})(6.00 \times 10^{14})$$

$$= 3.98 \times 10^{-19} \text{ J}$$

Example 2

A light source emits photons of red light of wavelength 700 nm while a radio antenna emits FM electromagnetic radiation of wavelength 3.00 m.

(a) What is the energy of a photon of the red light?

(b) What is the energy of a photon of FM electromagnetic radiation?

(c) Calculate the ratio $\dfrac{\text{Photon energy of red light}}{\text{Photon energy of FM electromagnetic radiation}}$.

Solution:

(a) The energy of a photon of red light is

$$E = \frac{hc}{\lambda}$$
$$= \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{700 \times 10^{-9}}$$
$$= 2.84 \times 10^{-19}\ \text{J}$$

(b) The energy of a photon of FM electromagnetic radiation is

$$E = \frac{hc}{\lambda}$$
$$= \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{3.00}$$
$$= 6.63 \times 10^{-26}\ \text{J}$$

(c) The ratio $\dfrac{\text{Photon energy of red light}}{\text{Photon energy of FM electromagnetic radiation}} = \dfrac{2.84 \times 10^{-19}}{6.63 \times 10^{-26}}$

$$= 4.28 \times 10^{6}$$

Example 3

A laser of output power 2.0 mW produces a monochromatic light of frequency 7.5×10^{14} Hz.

(a) Calculate the rate at which the photons are emitted from the laser.

(b) How will the rate of the photon emission change if the same laser were to produce light of lower frequency at the same power?

(c) If a laser which produces a monochromatic light of frequency 7.5×10^{14} Hz, can emit 1.0×10^{18} photons per second, what is its power rating?

Solution:

(a) Power $P = \dfrac{\text{Total energy } E}{\text{Time } t}$

If N number of photons are emitted in t seconds, the total energy E of these emitted photons is

$$E = Nhf$$

i.e. $P = \dfrac{Nhf}{t}$

Hence, the rate at which photons are released is

$$\frac{N}{t} = \frac{P}{hf}$$

$$= \frac{2.0 \times 10^{-3}}{(6.63 \times 10^{-34})(7.5 \times 10^{14})}$$

$$= 4.02 \times 10^{15}\ \text{s}^{-1}$$

$$= 4.0 \times 10^{15}\ \text{s}^{-1}$$

(b) Since P and h are constants, $\dfrac{N}{t} \propto \dfrac{1}{f}$. Hence, if frequency f is reduced, the rate of photon emission is increased.

(c) The power rating of the laser is

$$P = \left(\frac{N}{t}\right)hf$$

$$= (1.0 \times 10^{18})(6.63 \times 10^{-34})(7.5 \times 10^{14})$$

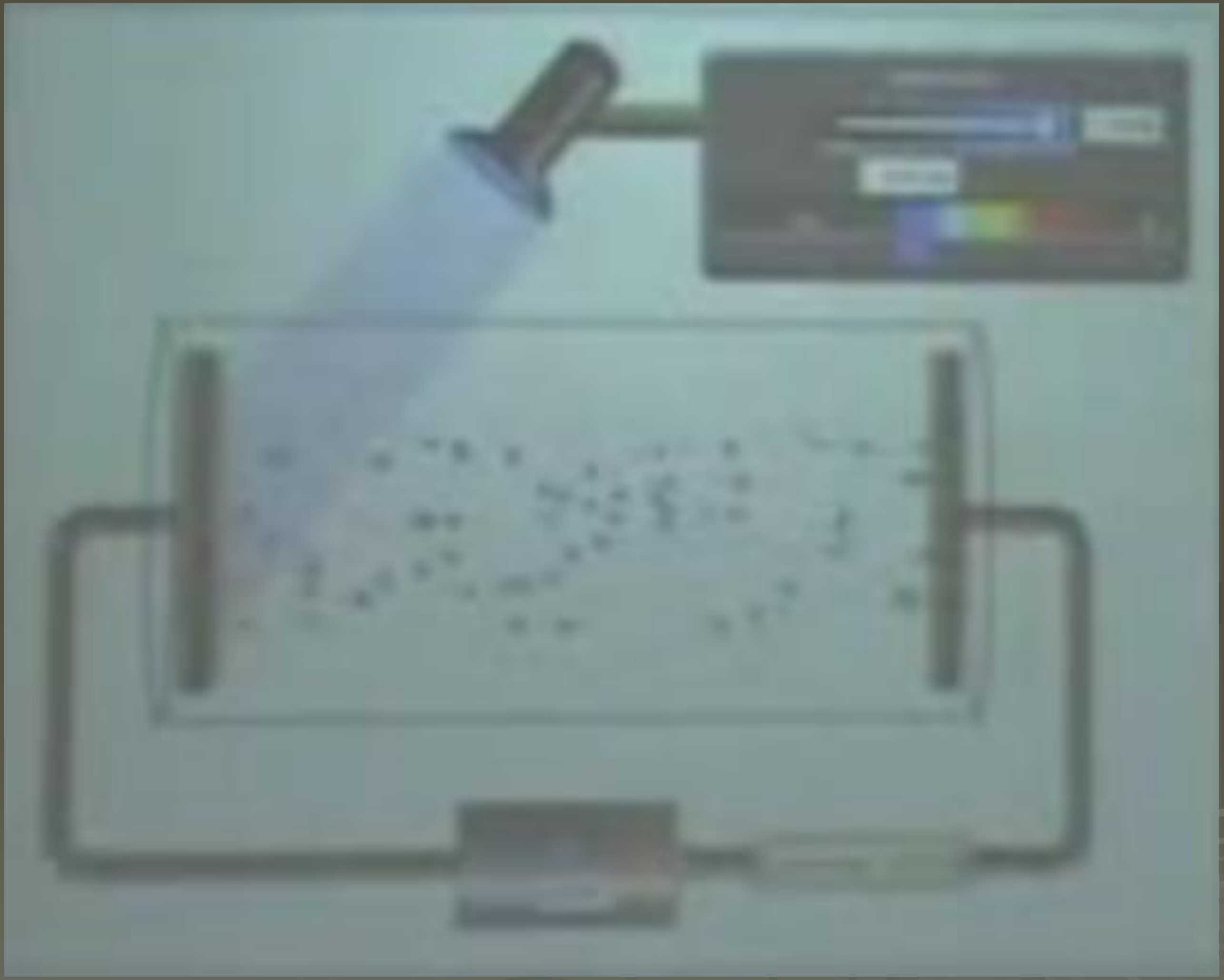
$$= 0.497\ \text{W}$$

$$= 0.50\ \text{W}$$

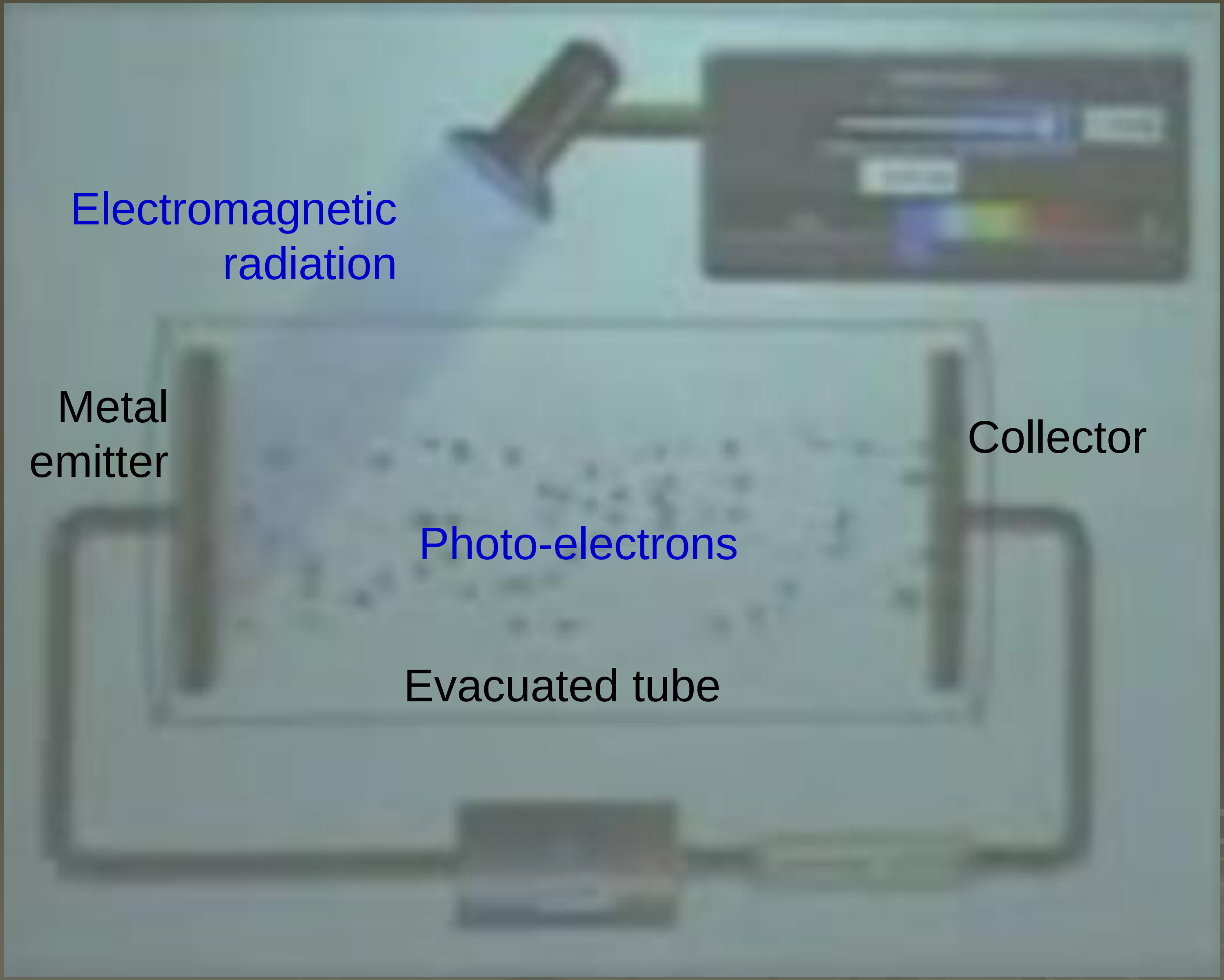
Photoelectric Emission

Photoelectric emission is the release of electrons from the surface of a metal when electromagnetic radiation is incident on its surface

The Photoelectric Effect



The Photoelectric Effect



The Photoelectric Effect

The experiment was first carried out by Einstein in 1905.

The observations / conclusions :

- If photoemission takes place, it does so instantaneously. There is no delay between illumination and emission
- Photoemission takes place only if the frequency of the incident radiation is above a certain value called the threshold frequency f_0
- Different metals need radiation of different threshold frequencies
- Whether or not emission takes place depends only on whether the frequency of the radiation used is above the threshold for that surface. It does not depend on the intensity of the radiation
- For a given frequency, the rate of emission of photoelectrons is proportional to the intensity of the radiation.

Explanation

Emission is instantaneous if above threshold frequency:

A single photon interacts only with a single electron. If the interaction is successful the entire energy *hf* of the photon is absorbed by the electron instantaneously and the photon ceases to exist.

The rate of emission of photoelectrons is proportional to the intensity of the radiation:

If *N* number of photons fall on the emitter in a time *t*, the intensity is

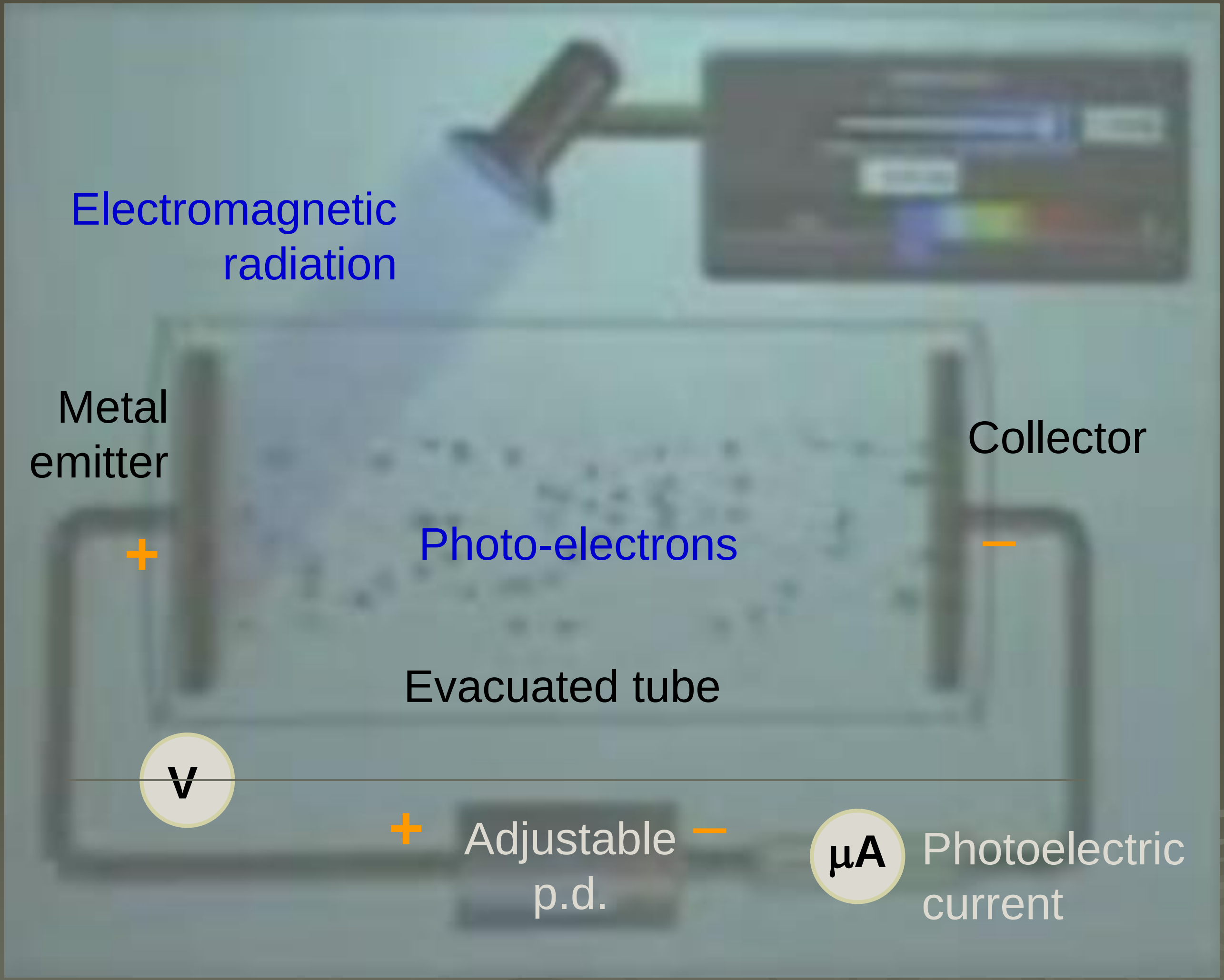
$$I = \frac{E}{tA} = \frac{Nhf}{tA} = \left(\frac{N}{t}\right)\frac{hf}{A}$$

Since f and A are kept constant:

$$I \propto \frac{N}{t}$$

The increase in the number of incident photons per unit time increases the number of photoelectrons as each photon emits an electron. Therefore the photoelectric current increases proportionally with the intensity of the radiation.

Kinetic Energies of Photoelectrons



Kinetic Energies of Photoelectrons

- The collector plate is made negative so that when the photo-electrons move towards it, they will lose their kinetic energies and gain potential energies

$$\frac{1}{2}mv^2 = eV$$

- The current flowing through the circuit is measured with a micro-ammeter and the potential difference between the emitter and collector measured with a voltmeter
- The voltage between the emitter and collector plates is gradually increased until the current drops to zero.
- The minimum value of the potential difference necessary to stop the electron flow is known as the stopping potential.
- If the experiment is repeated with radiation of greater intensity but same frequency, the maximum current in the micro-ammeter increases but the stopping potential is unchanged.

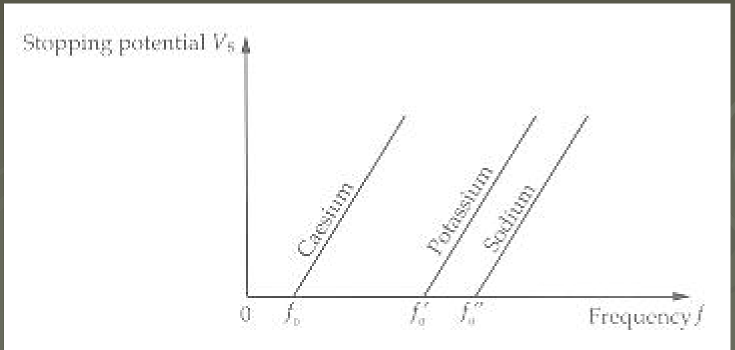
Explanation

If f or λ of radiation is kept constant, increasing the intensity of the radiation does not change the maximum kinetic energy of the photoelectrons and the stopping potential:

Increasing the light intensity simply increases the number of photons falling on unit area in unit time. This results in an increase in the emission of photoelectrons and the photocurrent. However, the incident photons still impart the same amount of energy hf to every electron.

If intensity I of radiation is kept constant but frequency f is increased, photoelectric current i remains constant but stopping potential increases:

Increasing the frequency increases the kinetic energy of the photoelectrons. It requires a larger stopping potential to reduce the photocurrent to zero



Work Function

- Threshold frequency f_0 is the minimum frequency of electromagnetic radiation that could emit photoelectrons from a material when the material is being irradiated.

- The existence of the threshold frequency suggests that electrons in the emitter are held weakly by electric forces within the material. In order to be ejected, the electron must absorb a certain amount of energy ϕ.

- We call this energy the work function of the material and it can be defined as the minimum energy necessary to remove an electron from the surface of the emitter material.

Einstein's Theory

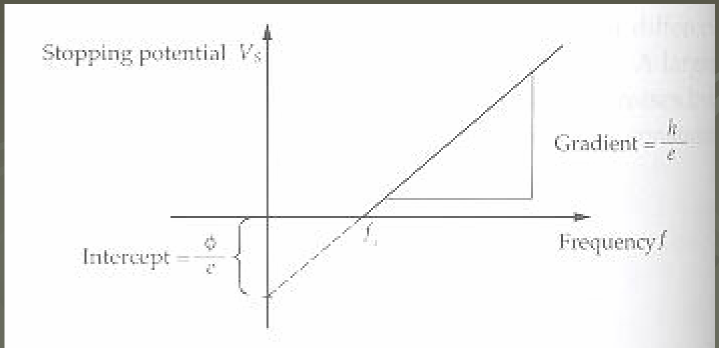
- A single photon has a quantum of energy hf. In a photoelectron interaction, the entire quantum hf is transferred to the electron in the emitter.
- The work function ϕ is dependent on the type of metal
- When a photon of threshold frequency f_0 is absorbed by an electron, the electron is released from the surface with zero kinetic energy. Therefore $hf_0 = \phi$
- When a photon of frequency f ($f > f_0$) is absorbed by an electron, it is released from the surface with a velocity that could range from the smallest v_{min} to the largest v_{max}.
- Therefore, Einstein's photoelectric equation is:

$$\boxed{hf = \phi + \tfrac{1}{2}mv^2 = \phi + eV_0}$$

V_0 = stopping potential

$hf / e = \phi / e + V_0$

$V_0 = (h/e)\, f - \phi / e$



Example 4

What is the maximum kinetic energy of a photoelectron ejected from magnesium when it is irradiated by an ultraviolet light of frequency 1.5×10^{15} Hz? [The work function $\phi = 5.9 \times 10^{-19}$ J.]

Solution:

Photon energy $= hf$

$$= (6.63 \times 10^{-34})(1.5 \times 10^{15}) \text{ J}$$

Maximum kinetic energy of the photoelectron is

$$
\begin{aligned}
K_{max} &= hf - \phi \\
&= (6.63 \times 10^{-34})(1.5 \times 10^{15}) - 5.9 \times 10^{-19} \\
&= 4.0 \times 10^{-19} \text{ J}
\end{aligned}
$$

Example 5

A conductor has a threshold wavelength of 0.85 μm. Calculate

(a) its threshold frequency f_0,

(b) its work function ϕ in electron volts and

(c) the maximum speed of the photoelectrons emitted when the conductor is irradiated by a blue light of wavelength 0.49 μm.

Example 5

(a) Threshold frequency f_0 is

$$f_0 = \frac{c}{\lambda_0}$$

$$= \frac{3.00 \times 10^8}{0.85 \times 10^{-6}}$$

$$= 3.53 \times 10^{14} \text{ Hz}$$

(b) Work function ϕ is

$$\phi = hf_0$$

$$= (6.63 \times 10^{-34})(3.53 \times 10^{14})$$

$$= 2.34 \times 10^{-19} \text{ J}$$

Since 1 eV $= 1.60 \times 10^{-19}$ J,

$$\phi = \frac{2.34 \times 10^{-19}}{1.60 \times 10^{-19}}$$

$$= 1.46 \text{ eV}$$

(c) From the photoelectric equation,

$$hf = \phi + \frac{1}{2} m v_{max}^2$$

$$v_{max} = \sqrt{\frac{2(hf - \phi)}{m}}$$

$$= \sqrt{\frac{2\left[(6.63 \times 10^{-34})\left(\frac{3.00 \times 10^8}{0.49 \times 10^{-6}}\right) - 2.34 \times 10^{-19}\right]}{9.11 \times 10^{-31}}}$$

$$= 6.14 \times 10^5 \text{ m s}^{-1}$$

The maximum speed of the photoelectrons emitted is 6.14×10^5 m s^{-1}.

Exercise 6

In a photoelectric experiment, a beam of light of wavelength 700 nm is incident on a metal. When the stopping potential is −1.5 V, no current is obtained. Calculate the work function ϕ of the metal.

Solution:

From the photoelectric equation,

$$hf = \phi + K_{max}$$

$$\frac{hc}{\lambda} = \phi + eV_s$$

$$\phi = \frac{hc}{\lambda} - eV_s$$

$$= \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{700 \times 10^{-9}} - (1.60 \times 10^{-19})(1.5)$$

$$= 4.4 \times 10^{-20}\ \text{J}$$

Example 7

The graph below (Fig 18.9) shows how the maximum kinetic energy K_{max} of the photoelectrons varies with the frequency f of the ultraviolet light that falls on metal A.

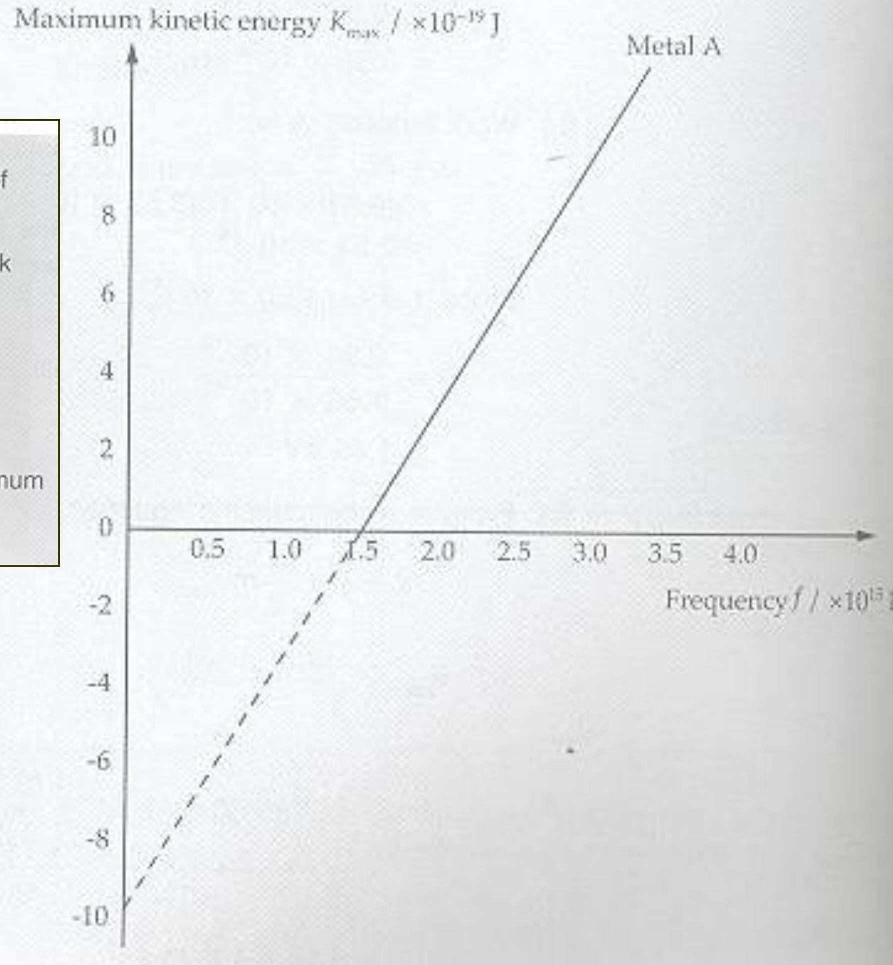
(a) Which aspect(s) of the graph could not be explained with the wave model of light?

(b) State the threshold frequency f_0 and hence, or otherwise, determine the work function ϕ of metal A.

(c) Use the graph (Fig 18.9) to calculate the value of Planck's constant h.

(d) What is the stopping potential V_s when ultraviolet light of frequency 3.0×10^{15} Hz irradiates on metal A?

(e) Using the same axis above, sketch a line to represent the variation of maximum kinetic energy K_{max} with frequency f for a metal with a larger work function. Label this as metal B.



▶ Fig 18.9

Solution 7

(a) The wave model of light could not explain why the maximum kinetic energy of the photoelectrons depends on the light frequency. It also could not explain why there should be a minimum (threshold) frequency f_0, below which no photoelectrons are emitted, irrespective of the intensity of light. It also could not explain why these photoelectrons can still be emitted instantaneously when light of very low intensity was used.

(b) Threshold frequency $f_0 = 1.5 \times 10^{15}$ Hz

Work function $\phi = hf_0$

$$= (6.63 \times 10^{-34})(1.5 \times 10^{15})$$
$$= 10 \times 10^{-19}\ \text{J.}$$

Alternatively, the work function ϕ could be read directly from the intercept obtained when the graph is extrapolated back to the maximum kinetic energy axis

i.e. $\phi = 10 \times 10^{-19}$ J

(c) From the photoelectric equation,

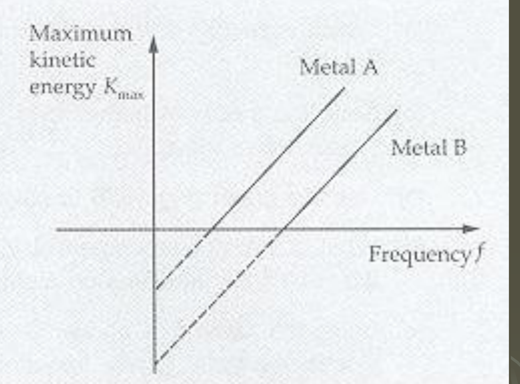
$$hf = \phi + K_{max}$$
$$K_{max} = hf - \phi$$
$$h = \text{gradient}$$
$$= \frac{[10 - (0)] \times 10^{-19}}{[3.0 - 1.5] \times 10^{15}}$$
$$= 6.7 \times 10^{-34}\ \text{J s}$$

(d) From the graph, when the frequency f is 3.0×10^{15} Hz, the maximum kinetic energy K_{max} is 10×10^{-19} J.

Since $eV_s = K_{max}$,

$$V_s = \frac{K_{max}}{e}$$
$$= \frac{10 \times 10^{-19}}{1.6 \times 10^{-19}}$$
$$= 6.3\ \text{V}$$

(e)



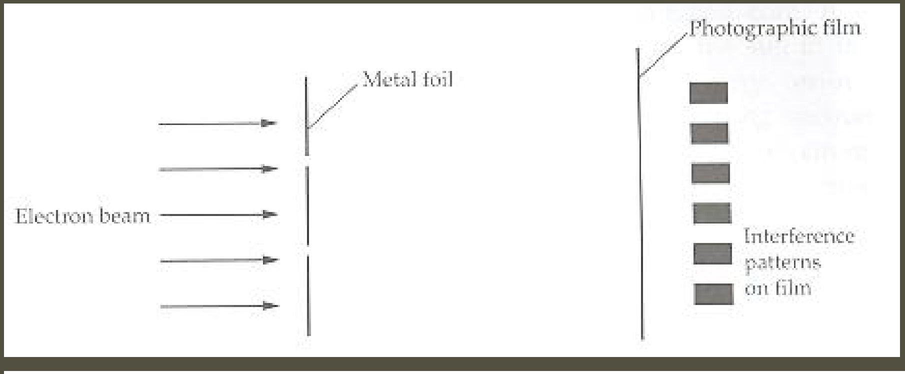
Wave-Particle Duality

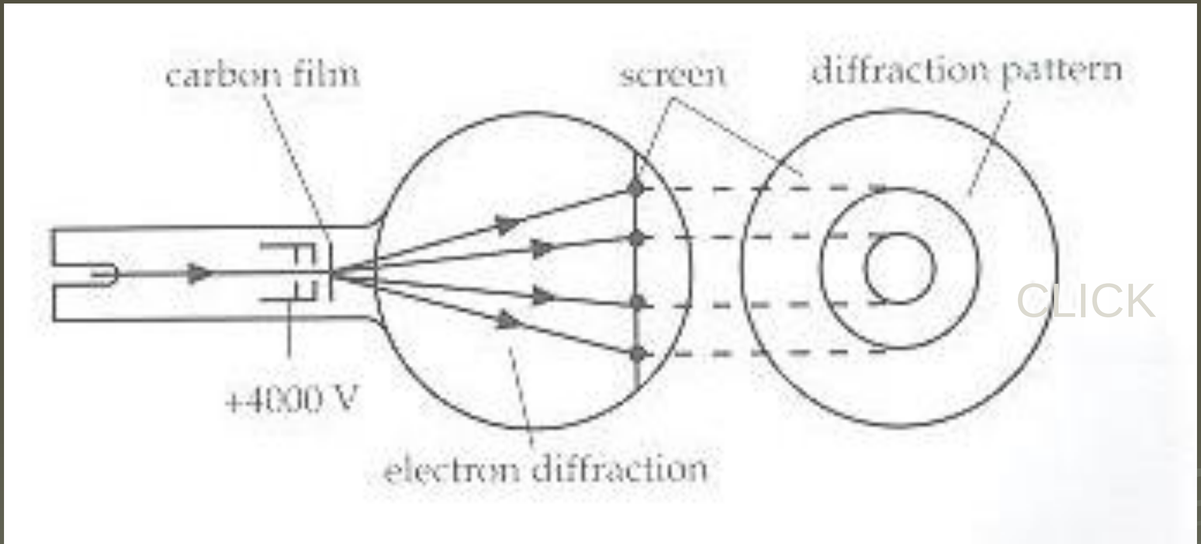
- The photoelectric effect shows that electromagnetic radiations have particulate nature.
- Observations show that moving particles also display interference and diffraction patterns which are wave properties
- De Broglie proposed that a particle with mass m and velocity v (for momentum p) also has wave properties

  De Broglie's wavelength: $\lambda = h / p = h / mv$

  Einstein theory of relativity: $p = E / c = hf / c = h / \lambda$

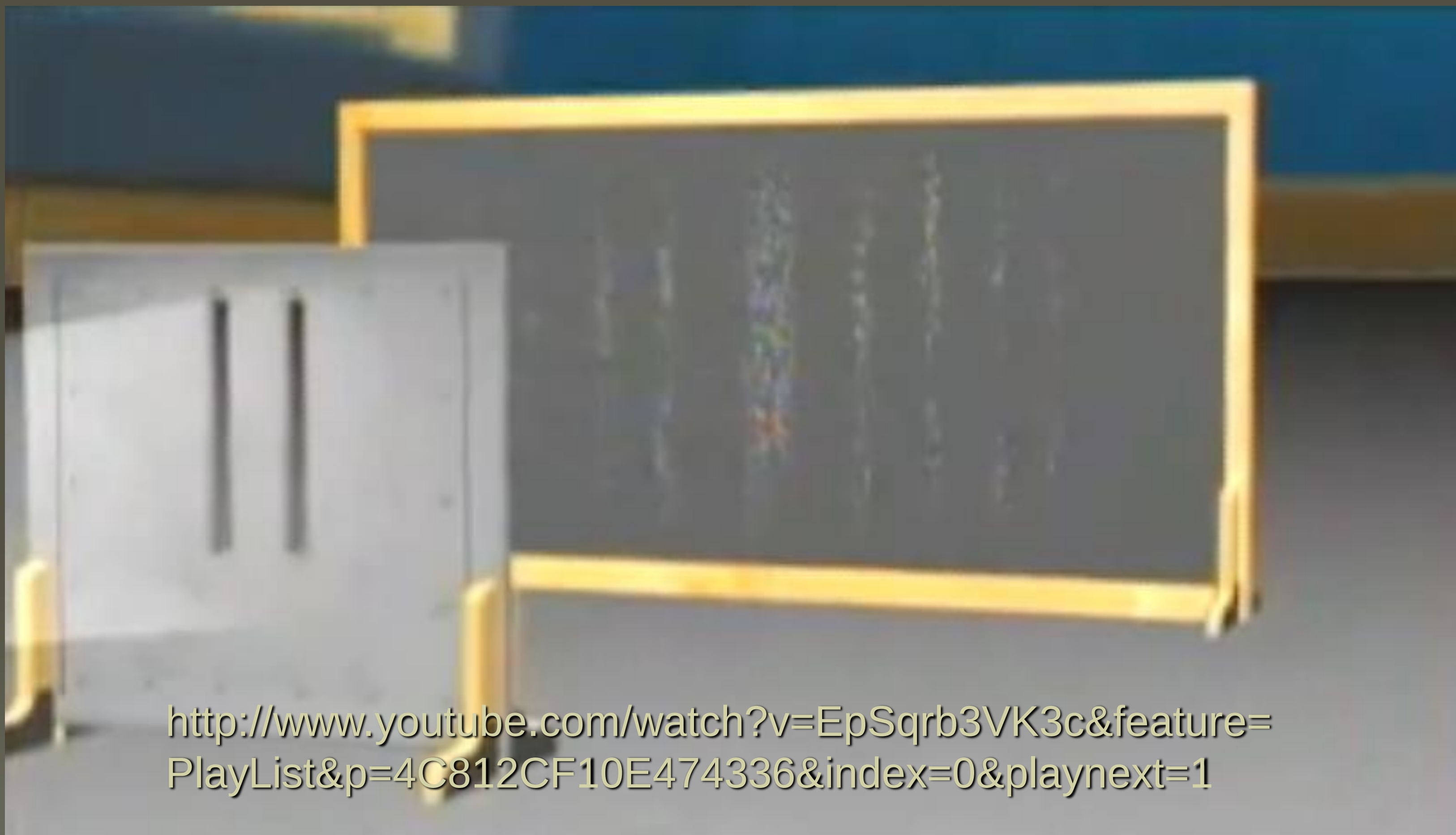
Wave-Particle Duality





Moving electrons display diffraction pattern.
Particles exhibit wave properties.

Wave-Particle Duality



http://www.youtube.com/watch?v=EpSqrb3VK3c&feature=PlayList&p=4C812CF10E474336&index=0&playnext=1

Example 8

Calculate the de Broglie wavelength of

(a) a 10 g bullet travelling with a speed of 300 m s^{-1}.

(b) an electron travelling with a speed of 6.0×10^6 m s^{-1}.

(c) a proton accelerated through a potential difference of 100 V.

(Mass of electron = 9.11×10^{-31} kg; mass of proton = 1.67×10^{-27} kg)

Solution 8

(a) From de Broglie's equation,

$$\lambda_{\text{bullet}} = \frac{h}{m_{\text{bullet}} v_{\text{bullet}}}$$

$$= \frac{6.63 \times 10^{-34}}{(0.010)(300)}$$

$$= 2.21 \times 10^{-34} \text{ m}$$

(b) From de Broglie's equation,

$$\lambda_{\text{electron}} = \frac{h}{m_{\text{electron}} v_{\text{electron}}}$$

$$= \frac{6.63 \times 10^{-34}}{(9.11 \times 10^{-31})(6.0 \times 10^{6})}$$

$$= 1.2 \times 10^{-10} \text{ m}$$

(c) From the law of conservation of energy,
gain in KE of proton = loss in PE of proton

i.e. $\frac{1}{2} m_{\text{proton}} v_{\text{proton}}^2 = eV$ where V is potential difference,

$$v_{\text{proton}} = \sqrt{\frac{2eV}{m_{\text{proton}}}}$$

Hence, $\lambda_{\text{proton}} = \frac{h}{m_{\text{proton}} v_{\text{proton}}}$

$$= \frac{h}{m_{\text{proton}} \sqrt{\frac{2eV}{m_{\text{proton}}}}}$$

$$= \frac{h}{\sqrt{2 m_{\text{proton}} eV}}$$

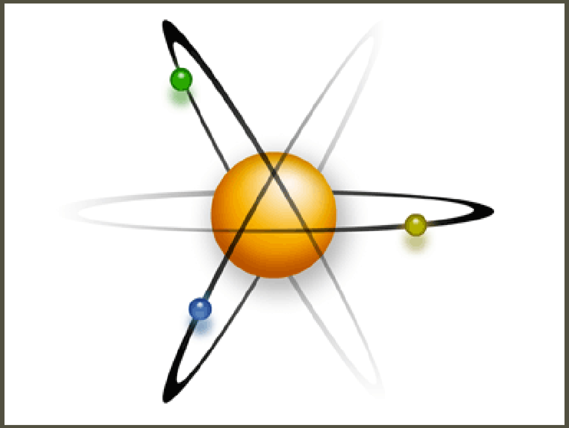
$$= \frac{6.63 \times 10^{-34}}{\sqrt{2(1.67 \times 10^{-27})(1.6 \times 10^{-19})(100)}}$$

$$= 2.9 \times 10^{-12} \text{ m}$$

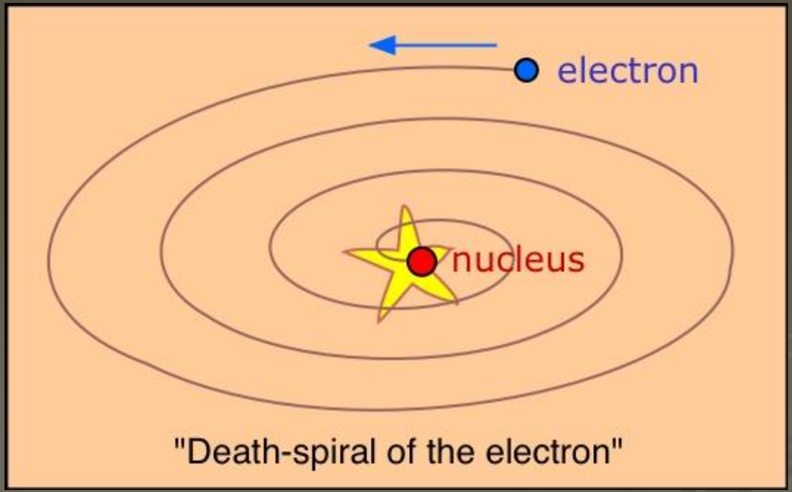
The Atomic Structure

Rutherford's Planetary Model:

In 1911, after the famous alpha scattering experiment, Rutherford proposed that electrons revolve at high speed in circular orbits around the positively charged nucleus.

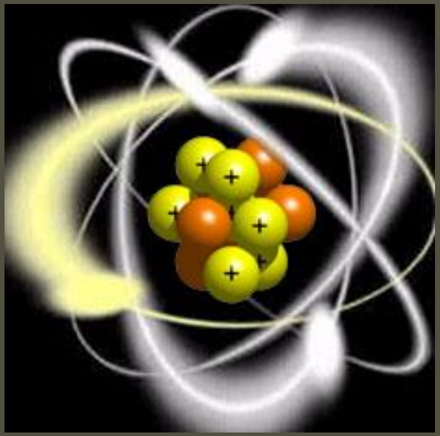


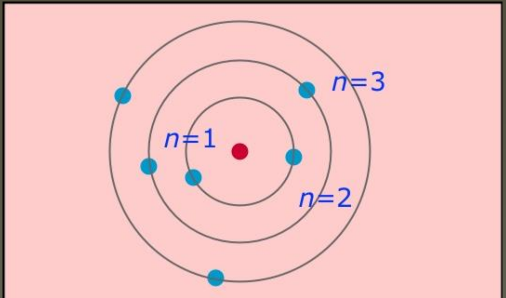
The Drawback:



According to classical electromagnetic theory, if a charged particle were accelerated around another charged particle then there would be a continuous radiation of energy. The loss of energy would slow down the speed of the electron and eventually the electron would fall into the nucleus. But such a collapse does not occur. Rutherford's model was unable to explain it.

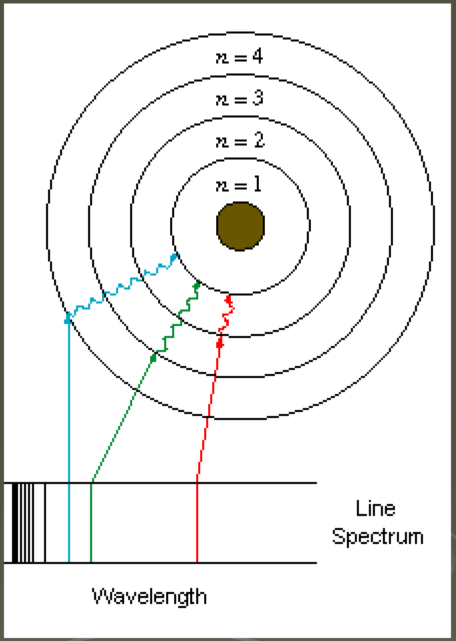
Neil Bohr's Atomic Model





In Bohr's model, electrons can only occupy orbits having integer values of *n*.

- **Niels Bohr**, in 1913 applies quantum theory to Rutherford's atomic structure
- He proposed that electrons travel in stationary orbits defined by their angular momentum.
- This led to the calculation of possible energy levels for these orbits and
- He postulated that the emission of light occurs when an electron moves into a lower energy orbit.



Bohr's Postulation

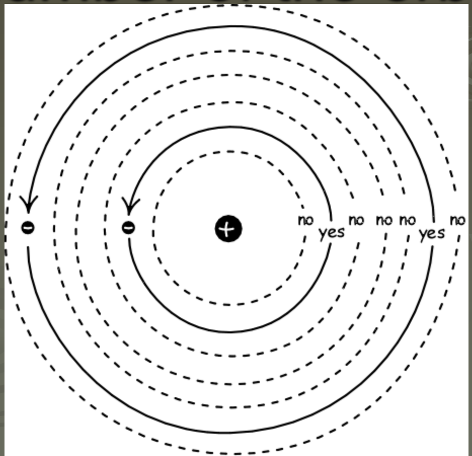
Electrons can move only in certain allowed orbits round the nucleus:

- Bohr's circular orbits are also called stationary states.
- Electrons in these stationary states behave very much like stationary waves fitted into the circumference of the orbit. This means: $2\pi r = n\lambda$

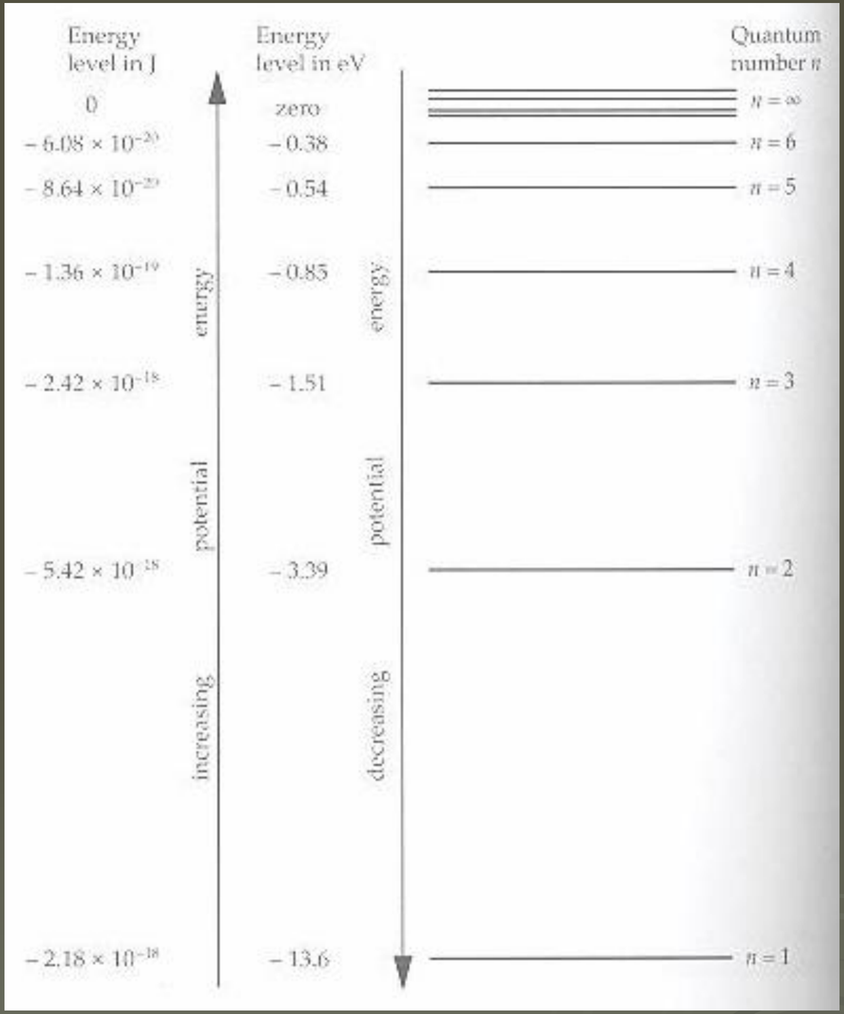
  where r is the radius of the orbit

  λ is the wavelength of the electron wave

  n is the integer called the quantum number of the orbit
- Electrons can exist in stationary states (n = 1, 2, 3 …) but not in between these states.



Bohr's Postulation



Energy levels for a hydrogen atom

Electrons in each orbit have a definite energy and they move in that orbit without radiating energy:

➢ A 'free' electron at $n = \infty$ has zero potential energy.

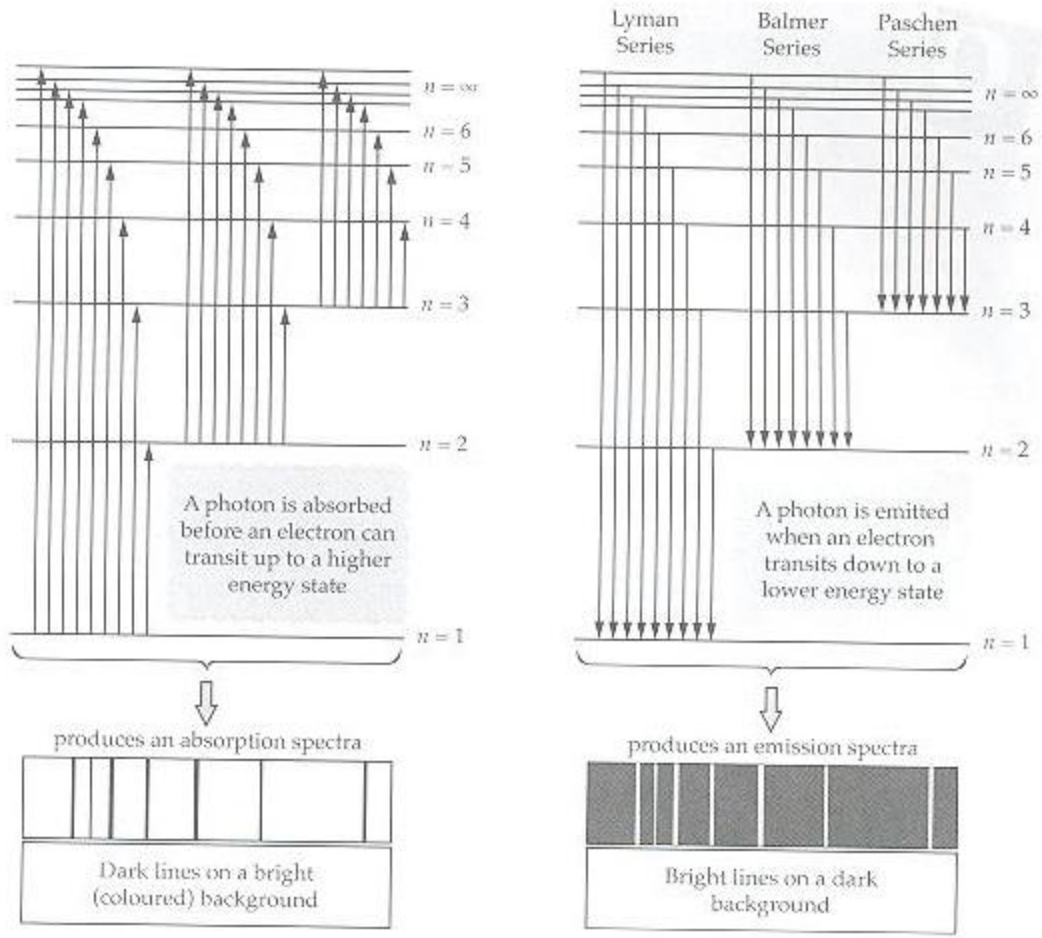
➢ All energy levels are negative indicating a loss of potential energy as the electron draws nearer to the nucleus.

➢ The energy level at $n = 1$ has the lowest potential energy and is called the ground state of the atom.

Ground state for H atom = -13.6 eV

➢ Historically, the quantum number n is called a shell where $n = 1$ is known as the K shell.

Electronic Transition



▲ Fig 18.14 Electron transitions from a lower to a higher energy state in a hydrogen atom

▲ Fig 18.15 Electron transitions from a higher to a lower energy state in a hydrogen atom

Electronic Transition

An electronic transition from one energy level to another requires the absorption or emission of a photon:

- This is represented by vertical arrows drawn between the energy levels.
- An electron must absorb energy before it can be excited from a lower to a higher energy level. This can be achieved in 3 ways:
  - An atom collides with an atom
  - An electron absorbs a photon of a certain frequency
  - An electron absorbs energy from a bombarding electron
- An electron in an upper energy level or excited state will fall back to a lower level after a short interval. This downward transition corresponds to the emission of a photon whose energy is the same as the energy difference between the levels.

$$hf = E_H - E_L$$

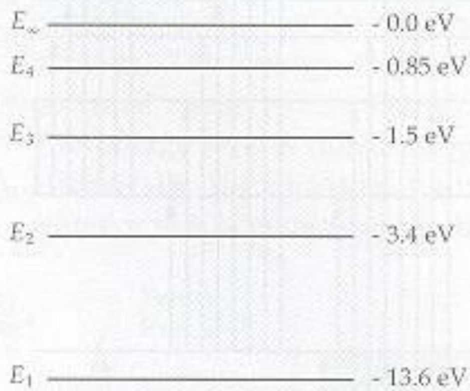
  where E_H = Energy of the upper state

  E_L = Energy of the lower state

- Energy difference between adjacent states are not equal.

Example 9

The first four energy levels of a hydrogen atom are shown below.



(a) What is the ionisation energy of the hydrogen atom?

(b) What is the minimum frequency of the photon that is required to cause an electron in the ground state of the hydrogen atom to be ionised?

(c) What would be likely to happen to photons of energies 12.75 eV and 15.0 eV if they pass through gaseous hydrogen atoms in the ground state?

(d) Bombarding electrons of energies 5.0 eV and 11.0 eV collide with gaseous hydrogen atoms in the ground state.

(i) How much energy is absorbed by the gaseous hydrogen atom electrons in the ground state?

(ii) How much energy is retained by the bombarding electrons?

(e) If a hydrogen atom is in the energy level of $n = 3$, show all possible transitions that the electron can undergo on the diagram and find the corresponding frequencies of the photons that are emitted or absorbed during the transitions.

Solution 9

(a) The ionisation energy of hydrogen atom $= E_\infty - E_1$

$= 0.0 - (-13.6)$

$= 13.6$ eV

(b) Minimum frequency for ionisation $f = \dfrac{E}{h}$

$$= \frac{13.6 \times 1.6 \times 10^{-19}}{6.63 \times 10^{-34}}$$

$= 3.3 \times 10^{15}$ Hz

(c) Photon of energy 12.75 eV:

$\Delta E_{14} = -0.85 - (-13.6)$

$= 12.75$ eV

The photon disappears as its energy of 12.75 eV is absorbed by the electron at E_1 and it subsequently transits to E_4.

Photon of energy 15.0 eV:

The photon will pass through the atom without being absorbed. The photon energy of 15.0 eV does not fit into any available energy intervals.

(d) For the bombarding electron of energy 5.0 eV:

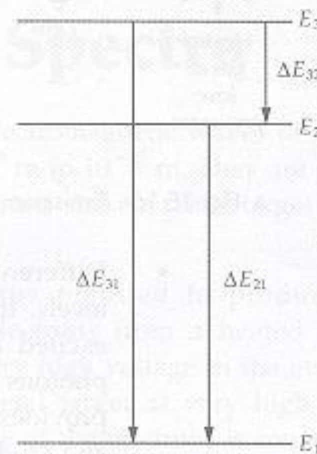
(i) The bombarding electron does not have enough energy to raise the electron in the hydrogen atom to the next level. Hence no energy is absorbed by the electron at the ground state.

(ii) The bombarding electron would retain most of its energy as it passes through the hydrogen atom.

For the bombarding electron of energy 11.0 eV:

(i) The electron at the ground state absorbs 10.2 eV from the bombarding electron and transits to E_2.

(ii) The bombarding electron leaves the hydrogen atom with 0.8 eV of energy.

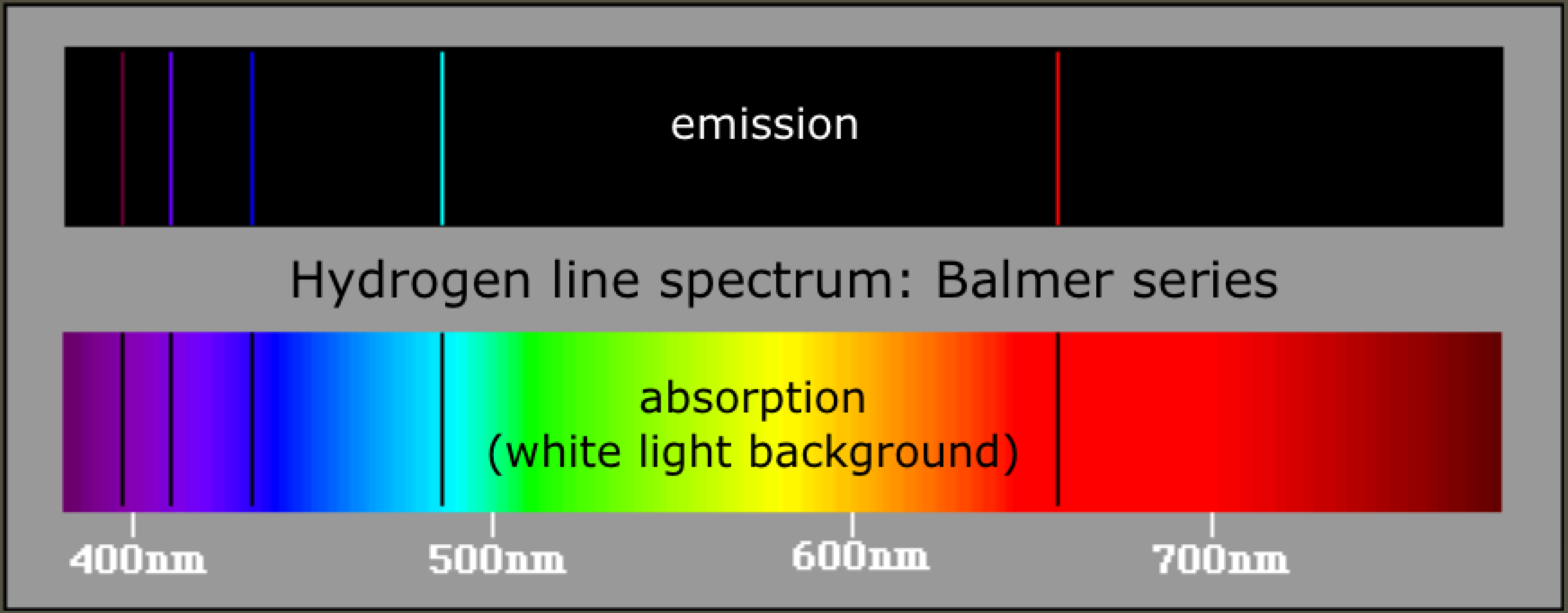


(e) $f_{31} = \dfrac{\Delta E_{31}}{h} = \dfrac{(12.1)(1.6 \times 10^{-19})}{6.63 \times 10^{-34}} = 2.9 \times 10^{15}$ Hz

$f_{21} = \dfrac{\Delta E_{21}}{h} = \dfrac{(10.2)(1.6 \times 10^{-19})}{6.63 \times 10^{-34}} = 2.5 \times 10^{15}$ Hz

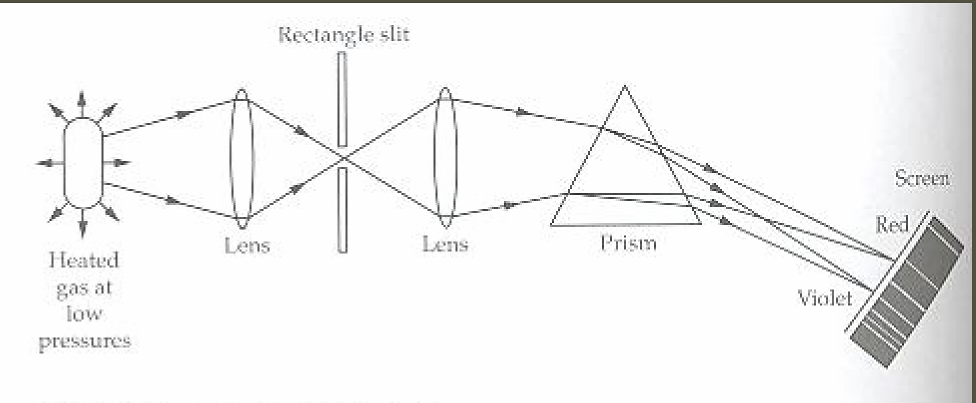
$f_{32} = \dfrac{\Delta E_{32}}{h} = \dfrac{(1.9)(1.6 \times 10^{-19})}{6.63 \times 10^{-34}} = 4.6 \times 10^{14}$ Hz

Line Spectra



- Emission line spectra are discontinuous coloured lines superimposed on a dark background.
- Absorption line spectra are discontinuous dark lines superimposed on a continuous spectrum of coloured lights.

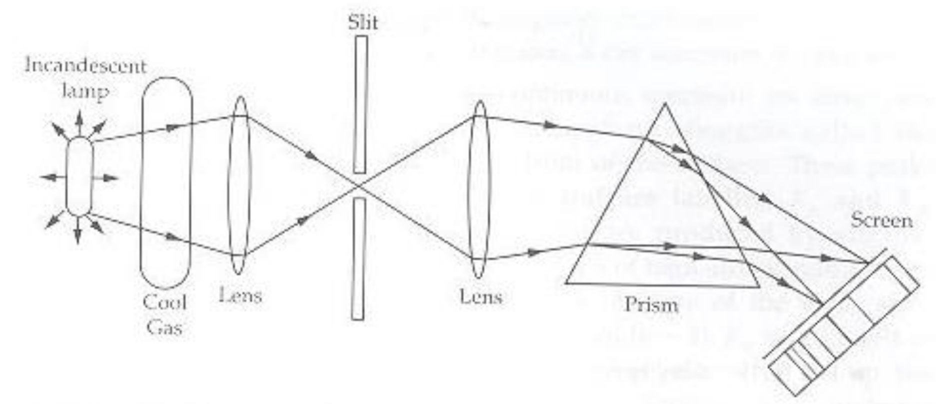
Emission Line Spectra



▲ Fig 18.16 Emission line spectra

- Emission line spectra are obtained by passing light, emitted from a heated gas at low pressure in a discharge tube through a spectroscope.
- The light is separated into different frequencies by a prism and brought to focus on different sections of a white screen by lenses.
- In total darkness, the screen will show coloured lines which are actually the images of the rectangular slit through which the light passes.
- Different elements have different configurations of energy levels. In a discharge tube at low pressure, electrons in their excited states fall back to lower energy levels by emitting photons of specific frequencies seen on the line spectra. This provides useful information for scientists to identify elements and study their atomic structure

Absorption Line Spectra



▲ Fig 18.17 Absorption line spectra

- Absorption line spectra are produced when white light from an incandescent lamp passes through a container of cool gas.

- Electrons from the ground state in the cool gas absorb some photons and transmit to the excited states.

- The continuous spectrum obtained on the screen has missing frequencies due to the absorption of photons by the cool gas.They appear as dark lines superimposed on a bright coloured background.

- The absorption line spectra can be observed in the continuous spectrum of the sun. The hottest central core emits white light but photons of some frequencies are absorbed by the cooler gases in the chromosphere (outer rim of the sun)

Physics is Great!

Enjoy Your Study!